Name: ____________________

UHID: ____________________

Homework
#06: Conditional Branching and Execution

Initialization

1. Create a constant string named **TESTDATA1** and give it the value "ECE3436" (Don't forget to terminate your string).

2. Create a constant vector named **TESTDATA2** with the elements 0,1,2,3 (all elements are 1 byte).
   .

3. Create a constant vector named **TESTDATA3** with the elements 4,5,6,7 (all elements are 2 bytes).

4. Create a 4-byte variable named **RESULT**.

Coding

5. Write the single instruction to ADD R2 to R3 and place the result in R1 only if the overflow flag is set (=1). Update the flags in the instruction.

6. Assume that register 0 (R0) has the value 0 and register 1 (R1) has the address of the constant string **TESTDATA1**. Write the code to count the number of "3"s in **TESTDATA1**. Place the result in R0.

7. Give the final value (in 32-bit hexadecimal) of register 0 and 4 after the following code is executed. Assume that all condition flags are clear (=0) before beginning execution.

R0 = ____________________

R4 = ____________________

```
        LDR   R0, =0x0
        LDR   R1, =0x1
        MOV   R2, #-1
        MOV   R3, #-2

        ADDS  R4, R0, R3
        BEQ   jump1
        SUBS  R4, R1, R1
        BMI   jump2
        MOVS  R4, R3
        BPL   jump3
        ADD   R4, R0, R1
        B     jumpEND

jump1
        MOV   R0, #0x1
        B     jumpEND
```

```
jump2
      MOV   R0, #0x2
      B     jumpEND
jump3
      MOV   R0, #0x3

jumpEND
      NOP
```

8. Convert the following C code snippets to ARM assembly. Assume all variables are 32 bits. You do not have to define the variables, but you must load/store their values from/to RAM. You are NOT allowed to use any Branch instructions for this problem

```
if(varX<varY) {
      varX=-1;
else
      varY=1;
}
```

9. Write the assembly code to calculate the dot product of **TESTDATA2** and **TESTDATA3**. Place the result in the variable **RESULT**. (Be sure to note the size of the different data elements.) You MUST use a loop to go through all elements in the vector. (The constants and variable have already been defined and initialized on page 1.)